

The Ephemeris

September 2018

Volume 29 Number 03 - The Official Publication of the San Jose Astronomical Association



September 2018 - December 2018 Events

Board & General Meetings

Saturday 9/22, 10/27, 11/17, 12/22,

Board Meetings: 6 -7:30pm

General Meetings: 7:30-9:30pm

Fix-It Day (2-4pm)

Sunday 9/9, 10/7, 11/4, 12/2

Solar Observing

Sunday 9/9, 10/7, 11/4, 12/2

Intro to the Night Sky Class & Houge Park 1Q In-Town Star Party

Friday 9/14, 10/19, 11/16, 12/4

Sights of the Cosmos Class & Houge Park 3Q In-Town Star Party

Friday 10/5, 11/30, 12/28

RCDO Starry Nights Star Party

Saturday 9/1, 10/6, 12/1, 12/29

Imaging SIG Mtg

Tuesday 10/16, 11/20, 12/18

Astro Imaging Clinics (at Coyote Valley OSP)

Saturday 10/6, 11/3

Quick STARt

Quarterly (by appointment)

Binocular Star Gazing

Will resume in Summer 2019

Please refer to the SJAA Website home page (www.sjaa.net) and click on any "Upcoming Event" highlighted in Blue. It will take you to the SJ-Astronomy Meetup page (www.meetup.com/SJ-Astronomy/events/) where you can find specific event times, locations, maps and possible cancellation due to weather.

INSIDE THIS ISSUE

President's Message	2
SSP Announcement	3
Solar Loaner Telescope	4-5
Bill O'Neil & Sandy Resignations	6
Cambrian Fest 2018	7
Articles	8-10
Astro Photos	11-13
Kid Spot	14
Board of Directors	15
Club Programs	16
Membership Form	17

Photo by SJAA Member: **Bruce Braunstein**

June 10, 2018

Location: Likely Place Golf & RV Resort in Likely California

President's Message September 2018

Dear Astronomers and Astronomy Enthusiasts,

First of all, let me welcome you all to the September edition of The Ephemeris. I hope you all enjoy reading this newsletter as much as we do putting it together.

Some of you already know me and have met me, but I expect a large number of members know nothing about me, except my name as the organization's President. So, let me share a few details about my astronomical journey with SJAA so far. About 8 years back, I accidentally stumbled upon SJAA while googling for star gazing opportunities in the Bay area. I went to RCDO for my first Starry Nights star party and met a bunch of very dedicated amateur astronomers. My daughter, who was then in middle school, was fascinated by all the telescopes and the breathtaking views of planets and clusters and galaxies that we got looking through the scopes. Soon after we attended the QuickSTART class, borrowed an 8" Dob from the Loaner program, and we were hooked! Rest, as they say, is history. Today, my daughter is pursuing Physics and Astronomy in college, and I am helping run this wonderful non-profit club that strives to bring astronomy and science to the general public. It is an honor to serve as the President of SJAA this year and to help the club achieve its mission!

Some of my friends and acquaintances who are not actively involved in Astronomy often express their wonder when I tell them anything about stars or planets or the deep sky objects. But when I talk to the very knowledgeable members of the club, I realize how little do I really know about space and our universe. So, for me, the beauty of this club is not just in the opportunity to go out and observe or image, but the opportunity to learn so much from my fellow members. That journey is just starting and will most certainly continue for the rest of my life. What an incredible storehouse of information our members are!

For this year, as the President of SJAA, I have a few initiatives in mind. One of them is to focus on automation of the various club operations, especially membership management. Another is to increase the member participation in the club operations. As all of you know, SJAA is entirely a volunteer run organization, and has no paid employees. However, despite having a few hundred members, only a handful are involved in actual operations of the club. I would like to urge all members - old and new - to get more involved with the club operations. We need all the help we can get!

This year, SJAA is also going through a few personnel transitions.

Bill O'Neil, our Vice President, and his wife Susan, are relocating out of the Bay Area to be close to their daughter. Bill has resigned from his positions of Director and Vice President. Bill has been with the club for many years - even he doesn't remember how many! He has contributed to the club in so many different ways that I perhaps can't even list them all. Not only has he been an active member and participant in many of our stargazing events, he has also been the program leader for the Starry Nights program and the Solar Viewing program in the past. He has been a Director for many years and has served as the Vice President since 2016. He was also the acting President last year during the then President's unexpected absence. Bill has also been involved in many different public outreach activities. During

my short few months as the President, I have come to rely upon Bill for invaluable advice and for being my backup whenever I was unavailable due to other commitments. I can't thank Bill enough for the incredible amount of work and research that he put in in finalizing the Reuse contract between SJAA and the City of San Jose. Bill, you will be sorely missed. But I wish you the best in your life away from SJAA. We hope you will come visit us often!

Another huge loss to the club is the resignation of Jim Van Nuland from the School Star Party program. Jim has been running this program for the last couple of decades and is a very well known name among many of the Bay area schools. He is practically a walking encyclopedia of all the information that one needs to know about conducting a star party at any Bay area school. Jim, I wish you the very best too in your life after retirement from SJAA.

Sandy Mohan is stepping down from her role of Content Editor of The Ephemeris. But she intends to stay involved with SJAA and I expect to see her around at the club events.

A very big "Thank You" to Bill, Jim, and Sandy for their years of contribution to SJAA.

I am also sad to inform you that Phil Chambers, our SCT expert in the FixIt program, passed away earlier this year after a brief illness. If anybody among our membership has any stories or pictures to share about Phil, I would love to hear from you.

Now, on the bright side, we have many new (and some older) people stepping up to take on various roles in the club. Carl Wong is the new lead for Starry Nights star party. Cat Cvengros is taking over as the new School Star Party lead from Jim. Jai Purandare is taking over the duties of the SJAA librarian. Jessica Johnson will be responsible for SJAA's social media presence. And Wolf Witt is taking over the Solar Viewing program from Bill, and has also started giving an introductory astronomy class called "Sights of the Cosmos". Please welcome Carl, Cat, Jai, Jessica, and Wolf in their new roles within SJAA.

With this, I will sign off for this edition. You will be hearing from me again in the future whenever I have any new or exciting developments to share. I do hope to meet each and every one of you at one (or more) of the SJAA events. And if you have any suggestions or questions, feel free to reach out to me at president@sjaa.net.

With best regards,
Swami Nigam

SJAA's School Star Party Needs Your Help!

Rob Jaworski

The San Jose Astronomical Association is an educational, science based non profit organization that has been operating in the south bay for over sixty years. Its mission is to bring the science of astronomy to the public by providing many different events and resources, most of which are free and open for anyone to experience. SJAA members set up telescopes at public parks for anyone to step up and get a close-up view the Moon or Jupiter. Astronomy talks, "Getting Started" classes and even telescope loans are examples of what the SJAA offers.

We are pleased to announce that we have a new leader of the School Star Party program! Cat Cvengros expressed interest in running the program, and she has already hit the ground running. She, along with her cohort Peter Mu, have met up with retiring program lead Jim Van Nuland, received all the information he had to give regarding how the program runs, and have brainstormed ideas for increasing the program's efficiency and updating its systems. They have reached out to the existing group of volunteer SSP volunteers and are looking for more. Be sure to keep an ear out for additional information, calls for volunteers and updates from Cat in the coming months. If you are interested in being a part of this most-important program of SJAA, please continue reading our original call out for volunteers. It provides the background of the program, its objectives and reach, and descriptions of how SJAA members can help. Huge thanks to Cat and Peter for taking on the important task of furthering the mission of the SJAA!

It's been said that astronomy is both a humbling and character building experience. In this age of hubris and arrogance, in today's world of screens both small and large, people, especially young people, may find it difficult to find examples of "humble". They may not know what it means to build character. But there is hope! The SJAA's School Star Party program partners with dozens of schools each year and touches hundreds of young people, students and their families and introduces them to the glorious sights of the night sky, its stars, constellations, planets and more. It's where people regularly see Saturn's rings or Earth's crusty, cratered Moon for the first time. They don't believe they are seeing it live, but when told it's the real thing, they shout with exclamations of joy. Every. Single. Time.

After 30 years of organizing and leading the SJAA School Star Party Program, Jim Van Nuland has retired. We now realize it will take a New Team to take over all the program aspects that Jim managed by himself.

SJAA is looking for a number of people who can help with this program. If you have a telescope or binoculars, or can explain very basic information about the night sky, or if you are good with organizing, we would love to hear from you. The School Star Party (SSP) program has several distinct roles that need to be filled, from fielding incoming requests for school star parties, to planning the event, or simply showing up and showing off the night sky. Read about them below.

If you are interested in learning more, please send an email to AskSJAA@sjaa.net.

There are four distinct roles, or areas of responsibility, for the SSP program. They can each be handled individually or two or more can be held by the same person. Here are their descriptions.

SSP Liaison

This role handles incoming requests for SSPs. When a teacher, parent or other member of a school community contacts the SSP program, the Liaison does the initial reply. In this reply, all the info that the member needs to know is provided. This includes things like how to determine which dates are good to have a star party, which upcoming dates are available and guidelines for having a successful school star party. Then internally, the Liaison starts working to identify a Scout who will 'own' this engagement with this school. Once the community member selects a few possible dates, the Liaison hands off the communication to the Scout with a message introducing her or him.

SSP Scout

The next role is that of the School Star Party (SSP) Scout. The SSP Scout owns the relationship with this community member and this school for the duration. The SSP Scout's duties include finalizing the SSP date out of the possible dates identified by the Liaison, then going to the school on some day before the SSP to survey the site, meeting the member and stakeholders in person, finding out if there are any special needs like insurance, and providing resources to help promote the event. At the same time, the SSP Scout identifies SSP Astronomers who can commit to the event. The Scout leads the event all the way through to the end of the event at that school.

SSP Astronomers

SSP Astronomers are the boots on the ground. They are the ones who get to hear the Oohs and the Ahhs from members of the school community. They are the on the ground liaisons between the SJAA, the community, and the wonders of astronomy. They provide their own scopes and related gear, share their knowledge, and they exercise their skills in clearly communicating to the general public. SSP Astronomers have to be able to commit to the date, communicate with the Scout, and work out their own transportation to and from the SSP, for themselves and their gear. The Scout does not necessarily have to be an SSP, but they do need to be at the event from start to finish, so it makes sense for her or him to bring a scope.

New Solar Loaner Telescope

Wolf Witt



Fig 1: Double-stacked Lunt 50 H- α Telescope (LS50THa with LS50C)

Good news, everyone! As of August, SJAA's Advanced Loaner Program has a hydrogen alpha solar telescope among its offerings.

This telescope is a pressure-tuned Lunt 50 H- α scope (model LS50THa), and the complete kit includes a second, front-mounted etalon filter (LS50C) for double-stacking, a projection sun finder, assorted eyepieces, a sun shade and a few other useful parts.

Why Hydrogen Alpha? Sunspots, Prominences and Filaments

Viewing the sun through our telescopes always requires special filters to ensure that the sun's energy does not damage our eyes. Standard telescopes equipped with white-light solar filters can provide nice views of sunspots and their structure. The sun, however, goes through a cycle of increasing and diminishing activity. The most recent greatest level of activity was recorded in 2014, and with a solar cycle period of roughly eleven years, we now find ourselves near the solar minimum, with the white-light sun often appearing as smooth as a cue ball. But in H-alpha light, the sun still offers some interesting views! While their frequency and intensity are also affected by the solar cycle, prominences and filaments are often awe-inspiring structures even around a solar minimum.

Prominences are eruptions of hydrogen plasma that are often loop or arc-shaped and extend beyond the limb (or edge) of the sun's disc. When a prominence occurs within the disc of the sun, it appears as a dark scar that is then referred to as a filament. White-light telescopes do not reveal these structures because the sun is so bright that the detailed patterns of plasma within the sun's atmosphere, the chromosphere, are drowned out. To see the rich structure of the chromosphere, we need a filter that discards most of the light of the sun and only passes light specifically associated with hydrogen. This (to us solar astronomers) precious wavelength of light is in the red part of the spectrum at 656.28nm; it is also referred to as the hydrogen alpha spectral line.

Pictures below show two examples of the sun in H-alpha light. While both of these pictures were taken with telescopes slightly larger than the 50mm loaner scope, the Lunt 50 can provide views that are nearly as detailed and certainly quite respectable.

The first picture (Fig 2) by Robert Duvall has been processed to enhance contrast and highlight detail. Because this picture was taken close to the last solar maximum, the left side shows some prominences that are extending nearly 100,000 miles into space. (Note the tiny earth image included for comparison.)

The second picture (Fig 3) was a single-frame photo taken in a hurry in 2017. It has not been post-processed, but a nice prominence and a large, vaguely W-shaped filament are nevertheless clearly visible on the left side.

Etalon Filters and Double Stacking

A filter that isolates only one spectral line from the entire color spectrum is a precision instrument. One incarnation of such an instrument is an etalon filter. This type of filter consists of a pair of partially reflective surfaces that form a resonant chamber within a light path. The coating of the reflectors and the spacing between the surfaces is tuned, such that the desired wavelength is reinforced and others are discarded. Hydrogen alpha telescopes have etalon filters that are designed to isolate that spectral line at 656.28nm.

A typical etalon filter in a commercial solar telescope has a pass band of 0.7Å or 0.07nm. This type of filter reveals prominences but filaments and other texture within chromosphere still tend to be elusive. To make a solar telescope even more discriminating, these filters may be double-stacked, meaning that a second etalon filter is added to the light path. The net effect is that the pass band narrows to 0.5Å. While that may not sound like much of a difference, the increase in detail within the sun's disc is often striking (although the brightness of some prominences may be slightly reduced).

But because few things are as easy as they sound, the etalon filters in our solar telescopes do pose a challenge. Changes in ambient temperature and pressure affect the position of an etalon's pass band, and even a fraction of a nanometer (e.g. the filter is centered at 656.50nm instead of the 656.28nm target) will greatly affect the visible detail. To compensate for environmental effects, users of etalon filters need to tune their filters to achieve good results. Consumer solar telescopes employ two different types of tuning mechanisms, pressure and tilt. The primary etalon filter that's permanently integrated into the Lunt 50 telescope is pressure tuned. The optional second filter for double-stacking is tilt tuned, so the SJAA solar loaner kit provides good opportunities for individuals to learn different tuning mechanisms and techniques.

For safety, note that etalon filters are necessary but not sufficient for a solar telescope. An etalon filter passes not only the target wavelength but also its harmonics. The energy in these harmonics will be outside of the visible spectrum but can still damage eyeballs. A safe and fully eyeball-friendly solar telescope also requires an energy rejection or blocking filter that throws away these unwanted harmonics. In the case of the Lunt 50, this blocking filter is in the diagonal (i.e. the Lunt B600 diagonal is not a standard star diagonal).

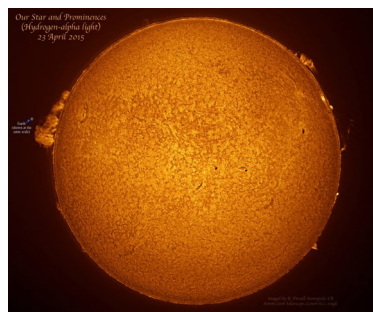


Fig 2: Detailed H-alpha Sun by Robert Duvall, April 2015; Lunt 80



Fig 3: Quickie H-alpha Sun by Wolf Witt, November 2017; Lunt 60



Fig 4: SJAA Lunt 50 Solar Loaner Kit

SJAA Solar Loaner Kit Details

The solar loaner kit comes in a stylish and portable yellow Pelican case and includes the following items (see also photo in Fig 4):

- Lunt 50mm H- α telescope (LS50THa) with B600 blocking filter and Sol Searcher projection finder
- Lunt 50mm H- α double-stacking filter (LS50C)
- Celestron 24-8mm zoom eyepiece
- Coronado CEMAX solar eyepieces
 - 25mm
 - 18mm
 - 12mm
- Coronado CEMAX Barlow lens
- Meade 4.5mm eyepiece (super magnification)
- Vixen dovetail rail
- Allen wrench tool (e.g. to attach dovetail to scope)
- Focuser locking screw (for use with heavy eyepieces or cameras)
- Dust blower bulb (not shown in photo)
- Sun shield (not shown)
- Documentation (not shown)

An individual who wants to borrow this kit need to provide her own mount, or arrange to borrow a mount through the SJAA loaner program.

The telescope may be attached to a mount via a standard camera tripod mounting screw or through a Vixen-style dovetail. The telescope tube's weight is specified at 7 lb and the double-stacking filter weighs in at 3 lb, so the mount needs to hold 10 lb plus eyepiece and sun shield in a comfortable and stable fashion.

Once equipped with the new SJAA Solar Loaner Telescope Kit and a suitable mount, you'll be able to enjoy views of the sun that relatively few people have seen. Enjoy!

Borrowing Procedure

The basic framework for the SJAA's Advanced Loaner Program is as follows:

- The typical loan period for every item is one month long (one loaner exchange day to the next).
- Standard loaner exchanges (pick up and drop off) occur once a month on a Sunday between 2PM and 4PM. Loaner exchange days typically align with Fix-It days. See sjaa.net calendar for details
- Requests for equipment must be received by midnight of the Thursday before the exchange day.

Furthermore, an individual who is borrowing the Lunt 50 solar telescope for the first time is expected to...

- Be a "qualified member" who has participated in an SJAA orientation program or has been interviewed and determined to have the skills and experience necessary to borrow this equipment.
- Arrive for pick-up at 1:30PM (i.e. 30 minutes earlier than the standard 2PM window).
- Complete a setup and use tutorial with a qualified representative of SJAA's solar program before 2PM.
- Share views through the loaner solar scope with the public during the Solar Sunday event that typically runs from 2PM to 4PM on the same Sundays as loaner equipment exchange.
- Return the telescope no later than 1:30PM on a subsequent loaner exchange day, so that the telescope becomes available to the next borrower in a timely fashion.

For complete information on SJAA's loaner program and to sign up, please visit:

<https://www.sjaa.net/programs/loaner-telescope-program/>

Links for Further Reading

LS50THa telescope:

<https://luntsolarsystems.com/product/lunt-ls50tha-50mm-pressure-tuned-h-alpha-telescope/>

LS50C filter:

<https://luntsolarsystems.com/product/lunt-ls50c-double-stack-filter/>

Solar Cycle:

<https://www.swpc.noaa.gov/news/solar-cycle-24-status-and-solar-cycle-25-upcoming-forecast>

Hydrogen Alpha Spectral Line:

<http://www.astronomyknowhow.com/hydrogen-alpha.htm>

Etalon Filters:

<https://luntsolarsystems.com/whats-an-etalon-and-how-do-i-make-one/>

<https://luntsolarsystems.com/pressure-vs-doppler-shift-part-1/>

<https://luntsolarsystems.com/pressure-vs-doppler-shift-part-2/>

Bill O'Neil Resignation Letter

Bill O'Neil

Bill's SJAA Resignation Letter July 28, 2018

I hope most of you have heard that Susan & I are taking advantage of this crazy real estate rental market and we are renting our Sunnyvale home for Big Bucks! (\$\$\$\$). On August 12, 2018, we will be moving up to Seattle Washington, to be near our daughter, Shannon, and her husband, Stephen, and possible future 'grandkid(s)'?

I bought my first "useless" Newtonian back in 2000 and bought a Celestron NexStar 5i with GPS, in 2003 (so Happy to see "Alignment Successful"). Not sure when I joined SJAA but started coming to Starry Nights and other Star Parties after Susan & I worked for part of the summer of 2013, as Volunteer Astronomer Park Rangers at Glacier National Park in Montana. Susan counted 4,500 visitors that looked through our telescopes in just four weeks in August.

I gave monthly "What's Up Tonight" talks at Starry Nights in 2014 and 2015. I took over as OSA Starry Night program leader for Chris Kelley in August 2015 and ran it until June of 2017, when Dave Ittner and later Carl Wong volunteered to be Docent leaders.

I worked with Michael Packer for years on the Solar Program after Michael convinced SJAA to invest in a 102mm Lunt, Hydrogen-alpha solar telescope back in February 2012. I took over the Solar Program from Michael in May of 2015. I'm very pleased to announce that Director Wolf Witt has 'volunteered' to run the Solar Observing, Education, & Outreach program for SJAA. He is already doing an excellent job! Thank You Wolf!

I joined the SJAA Board of Directors several years ago and was elected VP in 2016. I even was acting President for several months, when Teruo took ill in 2017. I was re-elected as VP this February and was able to convince Swami Nigam to serve as our new President.

Swami and I recently visited San Jose City Hall on July 16, 2018 and handed in our Signed CoSJ ReUse Contract for Houge Park (52 pages), which will last until December 2, 2019.

I really appreciate all the friends I've made at SJAA, and all the support while volunteering to do public outreach at Solar Parties and Starry Nights for SJAA. And the SJ Library(s) Total Solar Eclipse Training last July & August. (We had 300+ visitors at the Bascom Library).

The time has gone by too quickly. But there is always Change!

Therefore, I must now resign as a member of the SJAA Board of Directors, and also resign as your Vice President, effective immediately (OK, maybe Aug. 12, 2018). I plan to keep my SJAA membership active and will return to the Bay Area from time to time when I can.

You can always get in touch with me at my Bill.ONeil.MSEE@gmail.com e-mail address.

I'm confident SJAA has a strong leadership team and I wish you all success for the future.

Sincerely,

Bill O'Neil

Editor's note

Sandy Mohan

Dear SJAA friends,

I want to thank the Association for the opportunity to serve as the Editor of Ephemeris for the last 5 years. My relationship with the SJAA goes back all the way to 2000 when my husband and I along with my two sons moved to the Bay Area from New Jersey. My interest in astronomy started in my teenage years (doesn't everybody's?) but went dormant when I became a busy working mom. Commute, raising kids and homework became my life. It was my younger son who rekindled my interest when he started learning about planets in his second grade and became super interested in astronomy. Catalyzed by my son's interest I found that SJAA was right in the neighborhood. Also, the town we chose to live, Saratoga in the Bay Area, is blessed with particularly dark skies thanks to a city ordinance that restricts street lights.

We started attending the Star Parties at Houge Park and learnt how to star hop. With the fabulous loaner program we got the Signature Dob to take home with us. My sons and I spent countless nights becoming best friends with the scope and the night sky. My older son wrote an article for the Ephemeris in 2001 and even took a few classes on how to make your own telescope. Then my sons grew up, got interested in other things, and moved away but I was hooked to this hobby. There were days

when I couldn't wait for nightfall so I could go out and search for the next Messier object or the next double.

I remember when the Ephemeris was a simple black and white publication that would predictably appear in my mailbox. The newsletter has seen several changes since those early days. We debated if we should make it color, online or print or both, monthly or quarterly, etc. It was decided that the online version would have the beautiful color photographs, (too expensive for print), bylaws were changed to make the frequency quarterly, and keep the snail mail for those that wanted the printed version. As the membership grew, we started getting more observing reports and there were more club events to cover. To make it kid friendly we introduced the Kid Spot page with astronomy jokes causing adults to roll their eyes and the astronomy quiz that sometimes stumped the adults.

I have met several wonderful members over the years – mostly people whose voices I recognize in the dark. As I step down from my role, I wish the new Production Editor all the very best and look forward to reading future editions of the Ephemeris.

Sandy Mohan

August 2018

Cambrian Fest 2018

Rob Jaworski

For the ninth straight year, the SJAA hosted a booth, and lots of solar scopes, at San Jose's District 9 community fair called *Celebrate Cambrian*. This event took place on Sunday, 26 August 2018.



This area of San Jose is known as Cambrian Park, and features the Cambrian Community Center, which is situated on the grounds of the former Camden High School campus. SJAA volunteers maintain a presence at this annual event to contribute back to the community in which our organization has its base-camp at Houge Park.

As a member of this local community, I have spear-headed all of SJAA's participation in the event. It's essentially a solar viewing event, so Wolf Witt, our new leader of the Solar Program, again stepped up to help with the event. We showcased three solar



scopes, the club's flagship Lunt 100mm H-alpha scope, a recently donated Lunt 80mm, and Wolf's personal Celestron. The 80mm rig was outfitted with a MallinCam camera which was wired to an outdoor-rated flat panel display to more easily show visitors what they will see in the eyepiece of the other two scopes. And finally, we had Wolf's Lunt 60 on his Celestron NexStar mount.



The sun helped us by hosting a couple of interesting features. There was a fairly large prominence at the four o'clock position which showed up nicely on the monitor and in the eyepiece. The sun also displayed a nice filament across the face of its disk, again easily spotted in the scopes and through the camera. One of the posters of the sun that we had out there coincidentally mimicked what we could see in the eyepiece, with the prominence at the 4 o'clock position and the location of the filament. Talk about cooperation!



We had a great group of volunteers manning the scopes and the booth. Our new social media manager, Jessica Johnson (who recently got SJAA on Instagram!), was doing a great job guiding

the visitors through the scopes.

We had another young lady helping all day, a high school student who is in the process of forming a new astronomy club at her school. We had two more high school students manning the booth and helping kids and their parents learn about ultraviolet light by making bracelets out of UV-sensitive beads. The beads were a hit, with oohs and ahhs all around once the beads quickly developed color when exposed to sunlight!

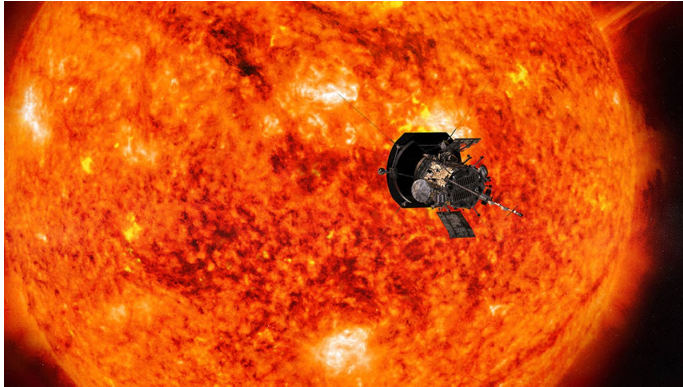


Bill Gottlieb also participated, helping explain aspects of what we were viewing. SJAA President Swami Nigam also paid a visit to ensure things were running smoothly.

Thanks to our volunteers who gave up most of their Sunday to help bring astronomy to the general public. We'll be out there again next year!

Traveling to the Sun: Why Won't Parker Solar Probe Melt?

Credits NASA



Artist's concept of NASA's Parker Solar Probe. The spacecraft will fly through the Sun's corona to trace how energy and heat move through the star's atmosphere.
Credits: NASA/Johns Hopkins APL

On August 12th, NASA's Parker Solar Probe launched to travel closer to the Sun, deeper into the solar atmosphere, than any mission before it. If Earth was at one end of a yard-stick and the Sun on the other, Parker Solar Probe will make it to within four inches of the solar surface.

Inside that part of the solar atmosphere, a region known as the corona, Parker Solar Probe will provide unprecedented observations of what drives the wide range of particles, energy and heat that course through the region — flinging particles outward into the solar system and far past Neptune.

Inside the corona, it's also, of course, unimaginably hot. The spacecraft will travel through material with temperatures greater than a million degrees Fahrenheit while being bombarded with intense sun light.

So, why won't it melt?

Parker Solar Probe has been designed to withstand the extreme conditions and temperature fluctuations for the mission. The key lies in its custom heat shield and an autonomous system that helps protect the mission from the Sun's intense light emission, but does allow the coronal material to "touch" the spacecraft.

NASA's Parker Solar Probe is heading to the Sun. Why won't the spacecraft melt? Thermal Protection System Engineer Betsy Congdon (Johns Hopkins APL) outlines why Parker can take the heat.

The Science Behind Why It Won't Melt

One key to understanding what keeps the spacecraft and its instruments safe, is understanding the concept of heat versus temperature. Counterintuitively, high temperatures do not always translate to actually heating another object.

In space, the temperature can be thousands of degrees without providing significant heat to a given object or feeling hot. Why? Temperature measures how fast particles are moving, whereas heat measures the total amount of energy that they transfer. Particles may be moving fast (high temperature), but if there are very few of them, they won't transfer much energy (low heat). Since space is mostly empty, there are very few particles that can transfer energy to the spacecraft.

The corona through which Parker Solar Probe flies, for example, has an extremely high temperature but very low density. Think of the difference between putting your hand in a hot oven versus putting it in a pot of boiling water (don't try this at home!) — in the oven, your hand can withstand significantly hotter temperatures for longer than in the water where it has to interact with many more particles. Similarly, compared to the visible surface of the Sun, the corona is less dense, so the spacecraft interacts with fewer hot particles and doesn't receive as much heat.

That means that while Parker Solar Probe will be traveling through a space with temperatures of several million degrees, the surface of the heat shield that faces the Sun will only get heated to about 2,500 degrees Fahrenheit (about 1,400 degrees Celsius).

The Shield That Protects It

Of course, thousands of degrees Fahrenheit is still fantastically hot. (For comparison, lava from volcano eruptions can be anywhere between 1,300 and 2,200 F (700 and 1,200 C) And to withstand that heat, Parker Solar Probe makes use of a heat shield known as the Thermal Protection System, or TPS, which is 8 feet (2.4 meters) in diameter and 4.5 inches (about 115 mm) thick. Those few inches of protection mean that just on the other side of the shield, the spacecraft body will sit at a comfortable 85 F (30 C).

The TPS was designed by the Johns Hopkins Applied Physics Laboratory, and was built at Carbon-Carbon Advanced Technologies, using a carbon composite foam sandwiched between two carbon plates. This lightweight insulation will be accompanied by a finishing touch of white ceramic paint on the sun-facing plate, to reflect as much heat as possible. Tested to withstand up to 3,000 F (1,650 C), the TPS can handle any heat the Sun can send its way, keeping almost all instrumentation safe.

Betsy Congdon of Johns Hopkins Applied Physics Lab is the lead thermal engineer on the heat shield that NASA's Parker Solar Probe will use to protect itself against the Sun. The shield is so robust, Congdon can use a blowtorch on one side and the other side remains cool enough to touch.

The Spacecraft That Keeps its Cool

Several other designs on the spacecraft keep Parker Solar Probe sheltered from the heat. Without protection, the solar panels — which use energy from the very star being studied to power the spacecraft — can overheat. At each approach to the Sun, the solar arrays retract behind the heat shield's shadow, leaving only a small segment exposed to the Sun's intense rays.

But that close to the Sun, even more protection is needed. The solar arrays have a surprisingly simple cooling system: a heated tank that keeps the coolant from freezing during launch, two radiators that will keep the coolant from freezing, aluminum fins to maximize the cooling surface, and pumps to circulate the coolant. The cooling system is powerful enough to cool an average sized living room, and will keep the solar arrays and instrumentation cool and functioning while in the heat of the Sun.

Continued on Page 8

Another issue with protecting any spacecraft is figuring out how to communicate with it. Parker Solar Probe will largely be alone on its journey. It takes light eight minutes to reach Earth — meaning if engineers had to control the spacecraft from Earth, by the time something went wrong it would be too late to correct it.

In the Astrotech processing facility in Titusville, Florida, near NASA's Kennedy Space Center, on Tuesday, June 5, 2018, technicians and engineers perform light bar testing on NASA's Parker Solar Probe. The spacecraft will launch on a United Launch Alliance Delta IV Heavy rocket from Space Launch Complex 37 at Cape Canaveral Air Force Station in Florida. The mission will perform the closest-ever observations of a star when it travels through the Sun's atmosphere, called the corona. The probe will rely on measurements and imaging to revolutionize our understanding of the corona and the Sun-Earth

The coolant used for the system? About a gallon (3.7 liters) of deionized water. While plenty of chemical coolants exist, the range of temperatures the spacecraft will be exposed to varies between 50 F (10 C) and 257 F (125 C). Very few liquids can handle those ranges like water. To keep the water from boiling at the higher end of the temperatures, it will be pressurized so the boiling point is over 257 F (125 C).

So, the spacecraft is designed to autonomously keep itself safe and on track to the Sun. Several sensors, about half the size of a cell phone, are attached to the body of the spacecraft along the edge of the shadow from the heat shield. If any of these sensors detect sunlight, they alert the central computer and the spacecraft can correct its position to keep the sensors, and the rest of the instruments, safely protected. This all has to happen without any human intervention, so the central computer software has been programmed and extensively tested to make sure all corrections can be made on the fly.

Launching Toward the Sun

After launch, Parker Solar Probe will detect the position of the Sun, align the thermal protection shield to face it and continue its journey for the next three months, embracing the heat of the Sun and protecting itself from the cold vacuum of space.

Over the course of seven years of planned mission duration, the spacecraft will make 24 orbits of our star. On each close approach to the Sun it will sample the solar wind, study the Sun's corona, and provide unprecedentedly close up observations from around our star — and armed with its slew of innovative technologies, we know it will keep its cool the whole time.

Local Author and Astronomer Honored with an Asteroid Named for Him

Bob Garfinkle

Former Union City Mayor, Councilmember, and longtime (30+ years) SJAA member Bob Garfinkle has received an out-of-this-world honor. In May 2018, Bob received word that the International Astronomical Union (IAU) has officially adopted the name "(31862) Garfinkle" for a Minor Planet (asteroid) originally designated 2000 EY70, which is located in the main asteroid belt between Mars and Jupiter. The object is about 6 km in diameter and has a visual magnitude of 20.5. Bob is also a Fellow of the Royal Astronomical Society, a life member of the Fremont Peak Observatory Association, and a best selling author of astronomy books. He is nearing completion of the proofing stage of his major lunar observers' handbook, which will be published by Springer in the near future.

Additionally, Bob is a former president of the California Writers Club (CWC) and in 2009 was a co-founder of his local Fremont Area Writers branch of CWC. He is also the

book review editor for the "Journal of the Association of Lunar and Planetary Observers," the Historian for the Lunar Section of the British Astronomical Association, and he writes articles on the history of astronomy for national and international publications.



Bob Garfinkle at Lick Observatory in 2016

Hubble's Messier Telescope



Charles Messier (1730–1817) was a French astronomer best known for his "Catalog of Nebulae and Star Clusters." An avid comet-hunter, Messier compiled a catalog of deep-sky objects in order to help prevent other comet enthusiasts from wasting their time studying objects that were not comets.

Credits: R. Stoyan et al., *Atlas of the Messier Objects: Highlights of the Deep Sky* (Cambridge University Press, 2008)

Although there are as many as one hundred billion comets in the outer regions of the solar system, prior to 1995, only around 900 had ever been discovered. This is because most comets are too dim to be detected without the proper astronomical equipment. Occasionally, however, a comet will sweep past the Sun that is bright enough to be seen during the daytime with the naked eye.

One such instance occurred in 1744. Comet Klinkenberg-Chéseaux, discovered by three amateur astronomers in late 1743, grew steadily brighter as it approached the Sun. By the end of February 1744, the comet had reached its peak brightness at an apparent magnitude of -7 , making it the brightest object in the sky except for the Sun and Moon. The comet's brilliance captured the interest of professional and amateur astronomers alike, including a young Charles Messier.

Born in 1730 in Badonviller, France, Messier had to give up formal education at age 11 when his father died. Soon after, he witnessed the spectacular Comet Klinkenberg-Chéseaux, which ignited his passion for astronomy. At the age of 21, Messier was hired as a draftsman for the French navy. He learned to use astronomical tools and became a skilled observer. For his efforts, Messier was eventually promoted to the chief astronomer of the Marine Observatory in Paris, where he pursued his interest in comets. He discovered over a dozen comets, earning him the nickname "Comet Ferret" from King Louis XV.

In 1758, Messier was in the process of observing one such comet when he was distracted by a cloudy object in the constellation Taurus. Upon further observation, he realized that the object could not be a comet because it was not moving across the sky. In an effort to prevent other astronomers from mistaking the object for a comet, Messier took note of it and began to catalog other comet-like "objects to avoid."

This comet-like object that Messier observed was NGC 1952. Commonly known today as M1 (Messier 1) or the Crab Nebula, it is the first object in Messier's *Catalog of Nebulae and Star Clusters*. By the time of his death in 1817, Messier had compiled a list of 103 objects in the night sky using his own observations with various telescopes and the discoveries of other astronomers. The catalog was revised in the 20th century and now contains 110 objects.

The Messier catalog includes some of the most fascinating astronomical objects that can be observed from Earth's Northern Hemisphere. Among them are deep-sky objects that can be viewed in stunning detail using larger telescopes but are also bright enough to be seen through a small telescope. This characteristic makes Messier objects extremely popular targets for amateur astronomers possessing all levels of experience and equipment. They are so popular, in fact, that they have inspired a special award from the Astronomical League (an organization for amateur astronomers) given to observers who are able to spot each of these objects. Those who succeed receive a certificate and are given the distinction of being in the Messier Club.

While the Hubble Space Telescope has not produced images of every object in the Messier catalog, it has observed 96 of them as of June 2018. Some of Hubble's photographs offer views of a given object in its entirety, but many focus on specific areas of interest. While Hubble is able to magnify objects very effectively, it has a relatively small field of view. This means that, in some cases, Hubble would need to take many exposures to capture an entire object. Although this is not always an efficient use of its time, as is the case for the widely spaced "open" star clusters in the Messier catalog, many exposures are taken when the scientific value justifies the time spent. One of these objects is the Andromeda galaxy (designated M31 in Messier's catalog). In order to create a mosaic image that depicts almost half of Andromeda, Hubble has taken nearly 7,400 exposures of the galaxy.

Unlike a digital camera that takes a single photograph in red, green and blue light to create a single full-color image, Hubble takes monochrome images at specific wavelengths of light. These specific wavelengths can reveal characteristics of an object that are of scientific interest, such as the presence of a particular chemical element. Multiple observations at different wavelengths can be combined to form a single image that reveals all of these characteristics at once but doesn't necessarily contain the full spectrum of visible light. In those cases, colors are assigned to each wavelength to highlight the different characteristics, offering a deeper understanding of the object's properties.

Additionally, Hubble is equipped to take infrared and ultraviolet images, which can reveal information that cannot be obtained using only visible light. Because infrared and ultraviolet light are not visible to human eyes, these images need to be processed in such a way that makes them meaningful to observers. This is done by assigning colors that humans can perceive to the wavelengths that they cannot.

Whether their tool of choice is a sophisticated ground-based telescope, a decent pair of binoculars, or simply their naked eyes, observers hunting for Messier objects can use the data gathered from Hubble's spectacular images to deepen their understanding of these 110 highlights of the night sky as they carry on the tradition of amateur astronomy.

The following images are some of the best from Hubble's Messier catalog.



M1



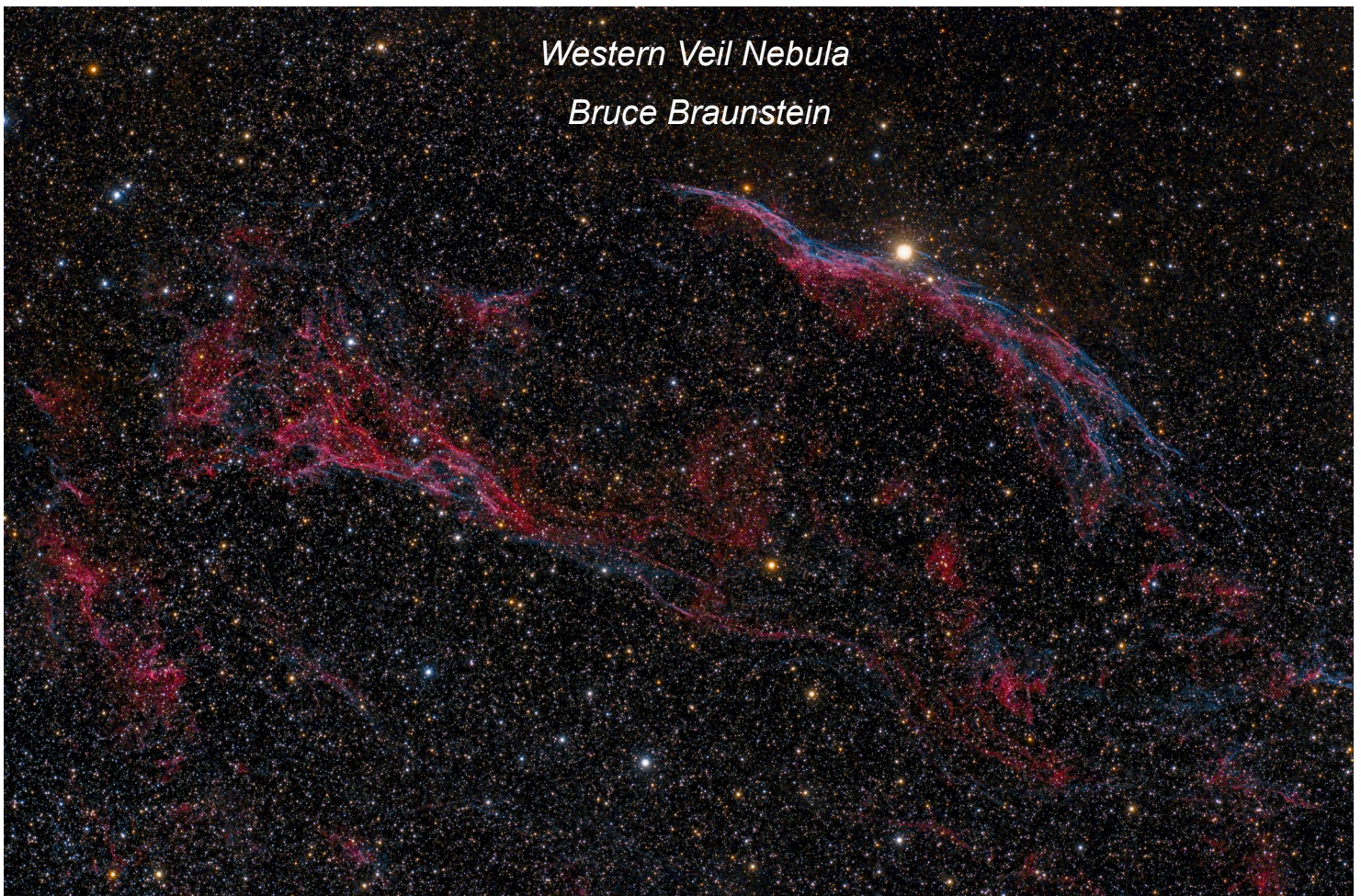
M2



M3



M4



Date: June 10th, 2018

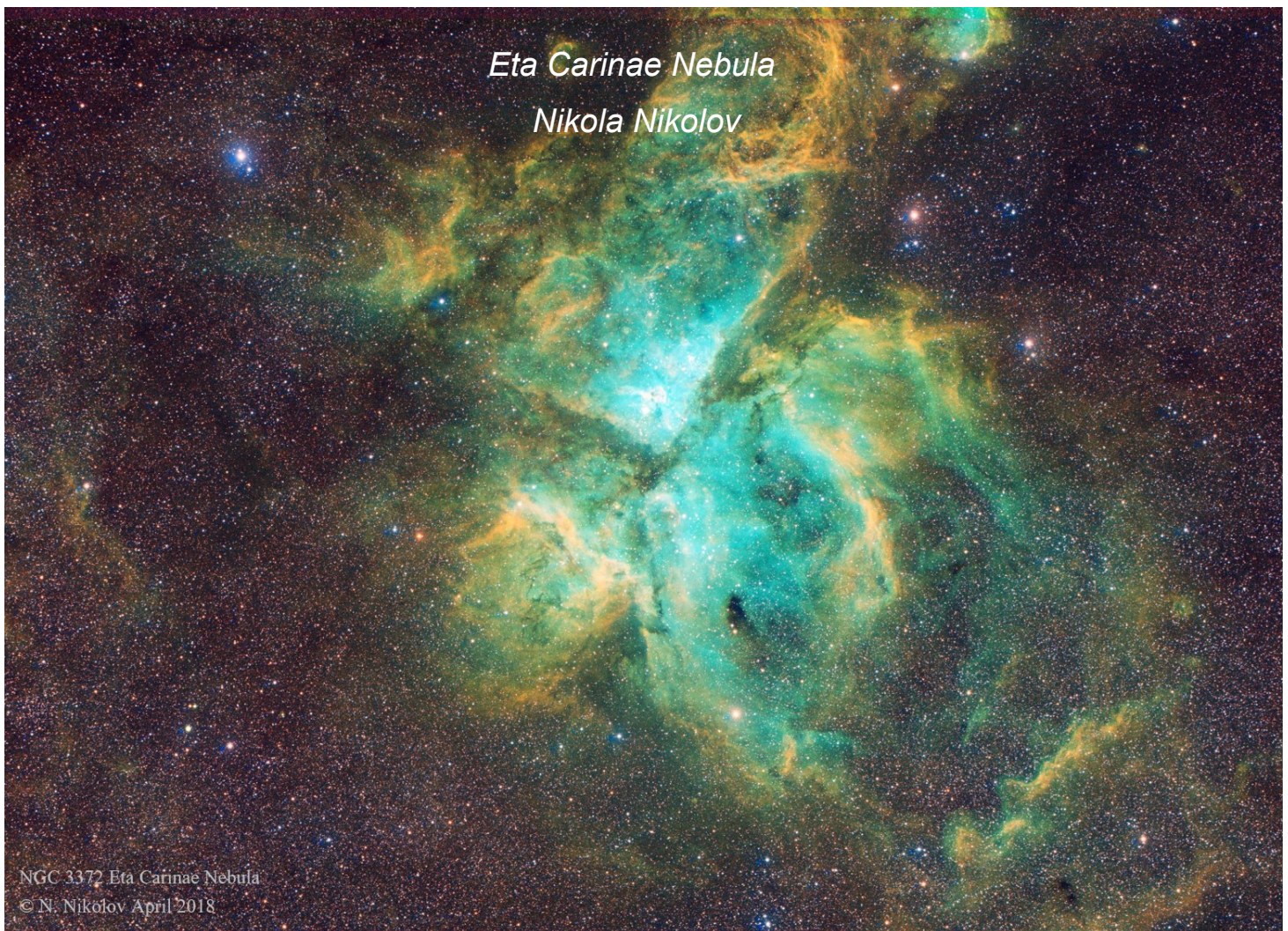
Location: Likely Place Golf & RV Resort in Likely California.

Telescope: Takahashi FSQ106 EDXIV Quad APO Refractor

Camera: QHY168C 25 exposures of 220 Seconds Each

Mount: Software Bisque Paramount MX+ - Unguided Using TPoint

Software: Software Bisque TheSkyX Pro, Sequence Generator Pro, Pixinsight, Photoshop CS6



Dates: 27 March - 02 April 2018

Location: Siding Springs Australia

Telescope: T12 Takahashi FSQ 106

Camera: CCD: FLI Microline 11002

Mount: Paramount PME

Software: AstroPixel Processor, MaximDL, Photoshop CS6

M. 16 The Eagle Nebula
Alan Pham

AP ALAN PHAM
ASTROPHOTOGRAPHY

Dates: May 26, 2018, June 1, 2018, June 7, 2018

Location: Home, San Jose, California

Telescope: Stellarvue SV4

Camera: ZWO ASI1600MM-Cool

Mount: Astro-Physics Mach1 GTO

Software: Sequence Generator Pro, PS, PixInsight, PHD2

Kid Spot



Kid Spot Jokes:

- ♦ **What is a light year?**
Same as a regular year but with less calories!
- ♦ **How does the solar system hold up its pants?**
With an asteroid belt!
- ♦ **What does a space turkey say?**
Hubble Hubble

Kid Spot Quiz:

1. Which of these objects is farthest from the sun?

a) Neptune b) Pluto c) 90377 Sedna

2. What is the visible part of the Sun called?

a) Lithosphere b) Photosphere c) Stratosphere

3. Approximately how many miles are there in a light year?

a) 9.3 billion miles b) 5.9 Trillion miles c) 5.9 Billion miles



Saving The Dark and SJAA Sukhada Palav

SJAA hosted a movie screening for Saving The Dark on June 30th. This movie is made by Sriram Murali, all on his own time and all with his own funds with a sheer passion for spreading awareness about light pollution. The movie is made in collaboration with International Dark Sky Association. It talks about how we are losing the dark to increasing light pollution and our future generations may not have any stars left to look at. It also touches upon how the light pollution affects different species and natural balance of the environment. Quite a few scientists appear in the movie including some amateur astronomers who describe the ways in which skies around us have changed.

The movie received a lot of applause and a long session of Q&A followed. SJAA has hosted a speaker from International Dark Sky Association before but the visual impact of this movie goes beyond what words can say. SJAA tries to do its part in supporting ventures like this. It was natural that one of our directors suggested officially sponsoring this movie, and subsequently the board approved the funding of one thousand dollars toward the costs associated with this film. I wish we knew about Sriram's project earlier and could participate more towards this cause but we are proud to be associated with it, even after the movie is made. SJAA will make a sincere effort to publicize the movie further so that more and more people watch it. That is how the awareness spreads after all. SJAA has renewed its membership in the International Dark Sky Association with a \$100 donation to the IDA.

Kid Spot Quiz Answers:

1. C
2. B
3. B

From the Board of Directors

Announcements

- ◇ Bill O'Neil is relocated to Seattle to be close to his daughter, and has resigned from his positions as Director and Vice President.
- ◇ Sandy Mohan is stepping down as the Ephemeris Editor
- ◇ SJAA Fall Swap Meet Sunday 28th October 12pm
- ◇ CalStar at Lake San Antonio September 5th - 9th

Board Meeting Excerpts

June 2, 2018

In attendance

Rob Jaworski, Bill O'Neil, Swami Nigam, Glenn Newell
Excused Absences: Sukhada Palav, Rob Chapman, Vini Carter, Wolf Witt
Guest: Emi Nikolov, Nikola Nikolov, Robert, Peter

Voting without a quorum

Swami made a motion such that for this meeting without a quorum, we will bypass non-critical topics, and anything that needs a vote before next board meeting, we will conduct voting via email per the bylaws.

Empty Board Seat

There were six applications for the open board seat. There were only two candidates that were eligible for this position. The two candidates were Ken Miura and Emi Nikolov

Upcoming Board Seat

Bill O'Neil has announced his intention to step down from the board, as well as his other positions such as VP, later in the summer of 2018. See Page 6 for additional information

City of San Jose Contract

Bill O'Neil has been reviewing it and the copy we received in May 2018 is what was unofficially agreed to last year, but was never signed/ratified. City also needs insurance related documents, and Bill will follow up with the insurance agent to get what's needed. .

Loaner Solar Telescope Update

Wolf has acquired a new 50mm Lunt Solar Telescope for the Loaner Program. He and Kanch are working on the Rules and Instructions. Wolf may need approval for out-of-pocket expenses for Solar.

June 30, 2018

In attendance

Rob Chapman, Bill O'Neil, Vini Carter, Glenn Newell, Sukhada Palav, Wolf Witt
Excused Absences: Rob Jaworski, Swami Nigam
Guest: Emi Nikolov, Cat C

Empty Board Seat

Ken Miura was selected by the board as new board member replacing Teruo's seat.

SJAA School Program Lead

Cat Cvengros attended the board of directors meeting as she indicated that she would be happy to help out with the school star party

Honorarium for Saving the Dark Movie screening

The movie producer has funded the production with his own money. Sukhada thought it would be reasonable to contribute more than our normal honorarium for the relatively high cost and effort that has gone into the production. A motion to offer an honorarium of \$300 to Sriram Murali for tonight's movie was approved.

July 25, 2018

In attendance

Sukhada Palav, Bill O'Neil, Vini Carter, Rob Chapman, Swami Nigam, Wolf Witt
Excused Absences: Glenn Newell, Rob Jaworski, Ken Miura
Guest: Emi Nikolov, Jessica Johnson, Peter Mu, Jai, Gary

Librarian

We have a new librarian: Jai Puranda. Sukhada is training her for the role

Understudy/Apprentices

Many of our programs are being run by only 1 or 2 members. And if the leader is not available for any reason, it really hurts the program. Swami proposed to build upon an idea floated in the past to recruit and train other members into apprentice/understudy type of role for the various club programs.

City of San Jose Reuse contract

The contract has been signed and hand-delivered to the city. We are now waiting for the city to approve and sign it, for it to become official.

Instagram

One of our new members, Jessica, would like to propose that SJAA develop a presence on Instagram. Jessica is willing to

manage an Instagram account for the club, as well as consider other social media such as Twitter and Facebook.

Sponsorship for Saving the Dark Movie

Sukhada proposed to replace the previously approved honorarium of \$300 by a sponsorship amount of a \$1000 in return for Sriram Murali adding SJAA's name in the movie credits as well as on the movie's website. The proposal was approved.

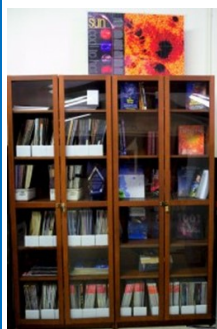
Bill O'Neil's resignation

Bill tendered his resignation from his positions as Director and Vice President because of his upcoming relocation to Seattle for family reasons. The board recognized Bill's as well as his wife, Susan's contributions to the club. We wish them all the best.

Star Party Request

Cat and Peter are hosting a STEM themed party in downtown SJ and would like to have a star party afterwards. Tues, October 23rd

SJAA Library



SJAA offers another wonderful resource; a library with good astronomy books and DVDs available to all of our members that will interest all age groups and especially young children who are budding astronomers! You may still send all your questions/comments

to librarian@sjaa.net.

Telescope Fix It Session

Fix It Day, sometimes called the Telescope Tune Up or the Telescope Fix It program is a real simple service the SJAA offers to members of the community for free, though it's priceless. Headed up by Vini Carter, the Fix It session provides a place for people to come with their telescope or other astronomy gear problems and have them looked at, such as broken scopes whose owners need advice, or need help with collimating a telescope.

<http://www.sjaa.net/programs/fixit/>

Solar Observing

Solar observing sessions, headed up by Wolf Witt, are usually held the 1st Sunday of every Month from 2pm - 4pm at Houge Park weather permitting. Please check SJ Astronomy Meetup for schedule details as the event time / location is subject to change.

<http://www.meetup.com/SJ-Astronomy/>

Quick START Program

The Quick START Program, headed up by Dave Ittner, helps to ease folks into amateur astronomy. You have to admit, astronomy can look exciting from the outside, but once you scratch the surface, it can get seemingly complex in a hurry. But it doesn't have to be that way if there's someone to guide you and answer all your seemingly basic questions.

The Quick Start sessions are generally held quarterly. The next Quick Start session during this time period is October 12th.

<http://www.sjaa.net/programs/quick-start/>

Intro to the Night Sky

The Intro to the Night Sky session takes place monthly, in conjunction with first quarter moon and In Town Star Parties at Houge Park. This is a regular, monthly

session, run by David Grover, each with a similar format, with only the content changing to reflect what's currently in the night sky. After the session, the attendees will go outside for a guided, green laser tour of the sky, along with views through SJAA Telescopes at the In-Town Star Party, to get a better look at the night's celestial objects.

<http://www.sjaa.net/programs/beginners-astronomy/>

Loaner Program

Muditha Kanchana (Kanch) heads up this program. The Program goal is for SJAA members to be able to evaluate equipment they are considering purchasing or are just curious about by checking out loaners from SJAA's growing list of equipment. Please note that certain items have restrictions or special conditions that must be met. If you are an SJAA member and an experienced observer or have been through the SJAA Quick START program please fill this form to request a particular item. Please also consider donating unused equipment.

<http://www.sjaa.net/programs/loaner-telescope-program/>

Astro Imaging Special Interest Group (SIG)

SIG has a mission of bringing together people who have an interest in astronomy imaging, or put more simply, taking pictures of the night sky. Run by Bruce Braunstein, the Imaging SIG meets roughly every month at Houge Park to discuss topics about imaging. The SIG is open to people with absolutely no experience but want to learn what it's all about, but experienced imagers are also more than welcome, indeed, encouraged to participate. The best way to get involved is to review the postings on the SJAA Astro Imaging mail list in Google Groups.

<http://www.sjaa.net/programs/imaging-sig/>

Astro Imaging Workshops & Field Clinics

Not to be confused with the SIG group this newly organized program championed by Glenn Newell is a hands on program for club members, who are interested in astrophotography, to have a chance of seeing what it is all about.

Workshops are held at Houge Park once per month and field clinics (members only) once per quarter at a dark sky site. Check the schedule and contact Glenn Newell if you are interested.

School Star Party

The San Jose Astronomical Association conducts evening observing sessions (commonly called "star parties") for schools in mid-Santa Clara County, generally from Sunnyvale to Fremont to Morgan Hill.

Contact SJAA's (Program Coordinator) for additional information.

<http://www.sjaa.net/programs/school-star-party/>

SJAA Contacts

President/Dir:	Swami Nigam
Vice Pres/Dir:	Vacant
Treasurer/Dir:	Rob Jaworski
Secretary/Dir:	Rob Chapman
Director:	Vini Carter
Director:	Ken Miura
Director:	Glenn Newell
Director:	Wolf Witt
Director:	Sukhada Palav
Ephemeris Newsletter -	
Editor:	Sandy Mohan
Prod. Editor:	Emi Nikolov
Binocular Stargazing:	Ed Wong
Fix-it Program:	Vini Carter
Hands on Imaging:	Glenn Newell
Imaging SIG:	Bruce Braunstein
Intro to the Night Sky:	David Grover
Library:	Jai Purandare
Loaner Program:	Muditha Kanchana
Memberships:	Dave Ittner*
Publicity:	Rob Jaworski
Quick START	Dave Ittner
Donations:	Dave Ittner*
Solar:	Wolf Witt
Starry Nights:	Carl Wong
School Events:	Cat Cvengros
Refreshments:	Marianne Damon
Social Manager:	Jessica Johnson
Speakers:	Sukhada Palav
Webmaster:	Satish Vellanki
E-mails:	http://www.sjaa.net/contact

*- acting

SJAA Ephemeris, the newsletter of the San Jose Astronomical Association, is published quarterly.

Articles, including Observation Reports for publication should be submitted by no later than the 20th of the month of February, May, August and November.

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243
<http://www.sjaa.net/contact>

Place
postage
here

San Jose Astronomical Association
P.O. Box 28243
San Jose, CA 95159-8243

Fold here

San Jose Astronomical Association Annual Membership Form

P.O. Box 28243 San Jose, CA 95159-8243

Membership Type (must be 18 years or older):

☐ **New** ☐ **Renewal**

☐ **\$20** Regular Membership with **online** Ephemeris

☐ **\$30** Regular Membership with **hardcopy** Ephemeris mailed to below address

The newsletter is always available online at: <http://www.sjaa.net/sjaa-newsletter-ephemeris/>

Questions? Send e-mail to: memberships@sjaa.net

Bring this form to any SJAA Meeting or send to the address (above). Make checks payable to "SJAA", or join/renew at: <http://www.sjaa.net/join-the-sjaa/>

Name:

Address:

City/ST/Zip:

Phone:

E-mail address:
